

LAFAYETTE 13 SE, LA, USA

WMO: 747600

Lat: 30.0918N

Lon: 91.8731W

Elev: 11

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 53960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	0.5	-8.0	1.9	0.9	-5.1	2.5	4.6	5.7	19.7	5.3	19.6	0.9	N/A	0.304

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	33.9	25.5	33.0	25.5	32.2	25.3	26.9	31.2	26.6	30.9	26.2	30.5	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	28.6	25.5	20.7	28.5	25.1	20.3	28.3	84.4	31.2	82.7	31.0	81.4	30.6	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.9	4.2	3.7	DB	-4.1	35.5	2.1	1.3	-5.6	36.4	-6.9	37.2	-8.1	37.9	-9.7	38.8	(4)
			WB	-5.9	27.8	2.4	0.4	-7.6	28.0	-9.0	28.3	-10.4	28.5	-12.1	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	11.2	13.3	16.8	19.8	23.7	26.7	27.3	27.4	25.5	20.8	15.6	12.5
	DBStd	6.99	5.56	5.27	4.43	3.51	2.80	1.44	1.24	1.48	2.13	4.04	4.93	5.35
	HDD10.0	129	53	25	6	1	0	0	0	0	0	0	10	35
	HDD18.3	817	228	153	82	25	3	0	0	0	0	23	109	193
	CDD10.0	3812	89	117	216	295	426	502	538	541	466	334	177	112
	CDD18.3	1461	7	12	34	70	171	252	279	282	216	99	27	12
	CDH23.3	12447	7	27	123	426	1399	2399	2655	2699	1757	782	136	37
CDH26.7	4536	0	1	3	41	412	984	1081	1147	653	203	10	0	
Wind	WSAvg	1.5	1.8	2.0	1.8	1.8	1.5	1.3	1.0	1.0	1.1	1.2	1.4	1.7
Precipitation	PrecAvg	1562	143	111	100	126	127	172	153	146	134	119	108	122
	PrecMax	2167	339	232	270	303	342	455	236	708	262	352	350	342
	PrecMin	1027	20	18	8	12	7	22	45	37	50	5	30	58
	PrecStd	288	92	60	60	74	86	94	47	122	60	99	75	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.3	25.8	27.0	29.1	32.6	35.1	34.6	35.5	33.6	31.6	28.0	25.8
		MCWB	19.8	21.1	20.8	22.4	23.4	24.9	25.6	25.5	25.2	23.6	21.9	21.9
	2%	DB	23.0	24.1	25.8	27.9	31.1	33.3	33.4	34.0	32.3	30.2	26.4	24.5
		MCWB	19.0	20.2	20.4	21.9	23.2	25.1	25.6	25.7	24.9	23.4	21.3	21.2
	5%	DB	21.5	22.8	24.9	26.9	30.1	32.3	32.6	33.0	31.4	29.0	25.0	22.8
		MCWB	18.6	19.9	20.1	21.2	23.2	25.1	25.7	25.9	24.6	22.7	20.3	20.1
	10%	DB	20.0	21.3	23.6	25.8	29.1	31.4	31.7	32.0	30.4	27.7	23.5	21.3
		MCWB	17.5	19.1	19.5	20.6	22.9	25.0	25.6	25.8	24.4	21.8	19.5	19.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	22.2	22.6	24.2	25.2	26.9	27.3	27.5	26.8	26.1	23.8	23.2
		MCDB	22.8	24.6	25.3	27.5	29.2	31.3	31.6	31.9	30.6	29.0	25.7	25.1
	2%	WB	20.5	21.5	21.7	23.5	24.7	26.4	26.8	27.0	26.1	25.3	22.6	21.9
		MCDB	22.1	23.6	24.4	26.4	28.6	30.8	31.1	31.3	30.1	28.5	24.7	23.7
	5%	WB	19.2	20.8	21.1	22.9	24.3	26.0	26.4	26.7	25.6	24.5	21.5	21.0
		MCDB	20.8	22.8	23.6	25.5	28.0	30.3	30.7	31.1	29.7	27.5	23.8	22.5
	10%	WB	17.5	20.0	20.4	22.2	23.9	25.6	26.1	26.3	25.2	23.6	20.4	19.8
		MCDB	19.4	21.7	22.8	24.8	27.6	29.9	30.2	30.7	29.1	26.2	22.8	21.2
Mean Daily Temperature Range	5% DB	MDBR	10.2	9.7	10.2	10.4	9.8	9.2	8.7	9.2	9.6	11.6	11.6	10.3
		MCDBR	9.9	9.4	9.4	10.1	10.2	10.5	9.8	10.4	10.2	11.0	11.1	10.0
		MCWBR	7.0	6.0	5.3	5.0	3.9	3.7	3.6	3.7	4.1	5.5	6.3	6.8
	5% WB	MCDBR	8.3	8.1	8.4	8.2	8.7	8.9	8.9	9.2	9.2	9.3	9.4	9.3
		MCWBR	6.8	6.2	5.6	5.2	3.7	3.8	3.7	3.8	4.1	5.3	6.5	7.5
Clear-Sky Solar Irradiance	taub	0.338	0.355	0.378	0.419	0.447	0.474	0.492	0.486	0.445	0.381	0.358	0.351	
	taud	2.505	2.466	2.389	2.278	2.238	2.207	2.191	2.207	2.289	2.445	2.454	2.474	
	Ebn at Noon	895	910	907	876	850	822	805	808	830	869	863	859	
	Edh at Noon	90	101	117	135	141	145	147	143	128	102	93	89	
All-Sky Solar Radiation	RadAvg	2.92	3.42	4.66	5.68	6.25	6.12	5.83	5.51	5.08	4.49	3.39	2.61	
	RadStd	0.28	0.50	0.41	0.41	0.36	0.50	0.42	0.49	0.45	0.52	0.32	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+80

Nomenclature: See separate page